

Siddhartha Reddy Munugala

New Jersey • (636) 262-1008 • Siddarth.lbf341@gmail.com • github.com/Redsid5 • linkedin.com/in/siddartharm • vibein.us • arenahq.us

PROFESSIONAL SUMMARY

Full-Stack Software & AI Engineer with experience architecting real-time systems, distributed databases, and generative AI pipelines. Co-founder of VibeIn/ArenaHQ, building and deploying production-grade web applications to 1,600+ users. Expert in TypeScript, Next.js, Node.js, Python, WebSockets, WebRTC, and PostgreSQL, focusing on scalable infrastructure and performance optimization.

EDUCATION

University of South Dakota New Jersey • Expected May 2027
B.S. Computer Science — Minor in Innovation & Entrepreneurship
Relevant Coursework: Data Structures, Algorithms, Systems Programming, Database Systems, Software Engineering

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL
Frameworks: Next.js 14, React, Node.js, Tailwind CSS
AI / ML: XGBoost, SHAP, PyTorch, Claude API, Gemini API, OpenAI API, LangChain, Whisper
Infrastructure: Supabase, PostgreSQL, Firebase, Neon Postgres, Redis, Docker, Vercel, Git, GitHub Actions
Other: WebRTC, WebSockets, REST APIs, FFmpeg, CodeMirror 6, tree-sitter, NetworkX

EXPERIENCE

Co-Founder & Lead Engineer — VibeIn / ArenaHQ Jan 2025 – Present
vibein.us • arenahq.us

- Architected and deployed two full-stack web applications using Next.js 14, Supabase, WebRTC, and Twilio, scaling user acquisition to 1,621 registered users with 20% active retention pre-beta.
- Debugged and resolved 8+ high-priority production issues—including Supabase RLS policies, real-time database locks, and API deprecations—under a hard deployment deadline to maintain 99%+ system uptime.
- Engineered high-performance real-time chat systems, peer-recommendation algorithms, and secure WebRTC P2P camera authentication pipelines, increasing user engagement and security validation.
- Streamlined continuous deployment pipelines using GitHub Actions and Vercel, reducing release cycles while maintaining 99.9% uptime across production environments.

Production Operations Staff — USD Athletics Aug 2025 – Present
University of South Dakota • Part-Time On-Campus

- Operated live-event audio systems and digital display infrastructure for NCAA Division I broadcasts under zero-tolerance real-time constraints, ensuring seamless production for televised games.

IT Coordinator & Publicity Chair — NSLS Executive Board Sep 2024 – Present
University of South Dakota

- Automated member onboarding pipelines and communication workflows using Python and custom dashboards, reducing manual administrative overhead by 50% for a 100+ member organization.

PROJECTS

EKO — AI Background Music & Voice Dubbing System 2026 – Present

- Designed a three-mode AI audio generation platform featuring automated multilingual vocal translation and scene-energy-matched music synthesis, improving audio production speed.
- Orchestrated a modern cloud stack utilizing Next.js 14, FastAPI, Neon Postgres, and Demucs stem-separation algorithms, enabling real-time audio isolation and processing.
- Optimized server memory footprints by 77% (from 1.8GB to 400MB) by migrating heavy audio model execution to the HuggingFace Inference API, resolving out-of-memory container crashes.
- Scaled the application to 821 registered users and 40% Daily Active Users (DAU) organically, drafting a technical system architecture preprint for arXiv.

PRISM — PR Risk Intelligence & Structural Modeling 2026

- Engineered an automated machine learning pipeline analyzing 45 source code features (AST complexity, dependency graphs, code churn) using XGBoost and SHAP to predict post-merge software regression risks with $AUC \geq 0.80$.
- Validated model generalization across heterogeneous codebases, maintaining a tight generalization gap ($\Delta AUC < 0.05$) and securing academic research endorsement.

COGN — Cognitive Drift Detection GitHub Action 2026

- Developed an open-source CI/CD GitHub Action that analyzes pull requests for semantic and structural drift introduced by AI generators, evaluating changes against codebase baselines to prevent technical debt.

ALASKA — Encrypted AI Assistant for Military Veterans 2026

- Architected an AES-256-GCM encrypted AI voice assistant integrating real-time crisis detection and automated veteran services routing, winning a top-10 placement out of 100+ teams in a 48-hour hackathon.

LEADERSHIP & ACTIVITIES

AI Education Workshop Facilitator Ongoing
Self-Initiated • Youth Outreach

- Designed and facilitated hands-on workshops teaching teenagers (ages 13–16) how AI integrates into healthcare, education, and media, reaching 50+ participants.
- Created accessible curricula covering basic neural network concepts and ethical AI frameworks, simplifying technical concepts for non-technical learners.
- Mentored teenagers in offline community spaces on foundational business, product development, and customer discovery principles, accelerating their communication confidence.
- Authored a 10-week strategic operational plan and built the chapter administration portal, reducing administrative overhead for the USD INFORMS Chapter.
- Coordinated campus tech events as IT Coordinator & Publicity Chair for the NSLS Executive Board and USD Marketing Club.